In-body Antigen Detection via Antibodies on a Transistor

CMPE 492 Final Report

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Date: June 2026
Repository: `https://github.com/duran0/CMPE492-Project`

Simulation-based feasibility study of HER2 transport, antibody binding, GFET current response, Debye screening, and design requirements.

Abstract

This report presents a simulation-based feasibility study for detecting Human Epidermal Growth Factor Receptor 2 (HER2) using an antibody-functionalized graphene field-effect transistor (GFET). The work extends the midterm diffusion and binding model into a staged final workflow that models HER2 transport, antibody binding, surface-bound molecule count, analytical GFET current response, limit of detection (LOD), Debye screening, and design requirements. The modeling chain is organized as transport $\rightarrow$ binding $\rightarrow$ transduction $\rightarrow$ noise/screening $\rightarrow$ detectability. A diffusion-only COMSOL Transport of Diluted Species baseline is compared with a partially anatomical prescribed-velocity convection-diffusion extension. Directional lymph-node-inspired flow improves HER2 exposure at the sensing region, and sensor placement affects response timing and current response. However, electrical detectability is conditional. It depends on transduction coupling, electronics noise, binding affinity, receptor-channel distance, and ionic screening. The main negative result is that PBS-like Debye screening strongly suppresses the effective GFET signal for longer receptor distances. The final output is therefore not experimental validation of an in-body detector, but a design requirement map showing the physical and electronic conditions under which the simulated GFET-HER2 concept could remain detectable.

Acknowledgments

I thank my advisors Tuna Tugcu and Kutlu Ulgen for their guidance on the modeling direction and final deliverables. I also thank M. Gorkem Durmaz, whose previous graphene-based biosensor modeling work provided useful conceptual guidance for this project.

Contents

Chapter 1

Introduction

1.1 Motivation and Broad Impact

Cancer develops when normal control of cell division fails, allowing malignant cells to invade surrounding tissue and spread through blood or lymphatic pathways. Breast cancer is one of the major clinical settings where molecular biomarkers influence prognosis and treatment selection. HER2 is especially important because overexpression or amplification of the ERBB2 gene can drive aggressive tumor behavior and guide therapy decisions [1–4].

The midterm stage established the biological and electronic motivation for a HER2-sensitive GFET biosensor. The final work keeps that motivation but narrows the claim. A deployable in-body detector is not demonstrated here. The project instead asks whether a physically consistent simulation chain can identify the main barriers between HER2 arrival and an electrical GFET signal. This framing is important because a sensor placed in a tissue-like environment does not directly measure the bulk antigen concentration. HER2 must first reach the sensing region, bind to the functionalized surface, create an effective charge perturbation, and exceed the current noise floor after electrostatic screening.

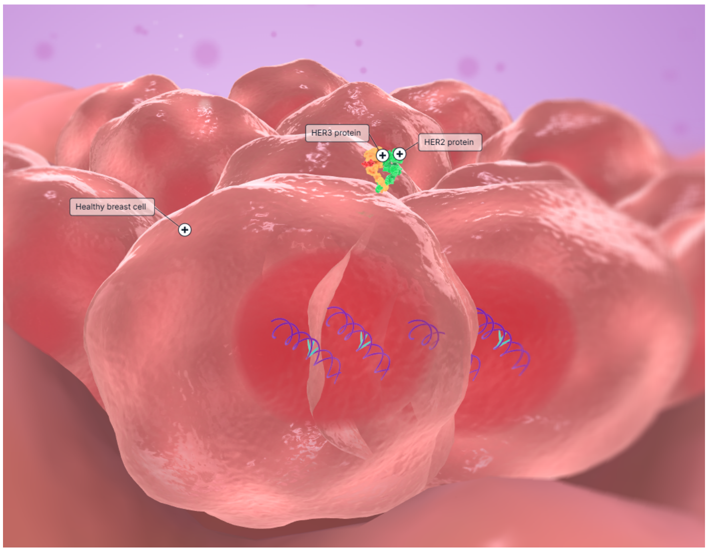

Figure 1.1: Healthy breast cell visualization showing the biological context for HER2/HER3 signaling [5].

1.2 HER2 as a Breast Cancer Biomarker

HER2 is a transmembrane receptor in the epidermal growth factor receptor family. In normal tissue, HER2 participates in signaling pathways related to proliferation and repair. In HER2-positive breast cancer, overexpression can increase growth signaling and alter clinical management [1, 2]. This project treats HER2 as the target antigen for a simulation-based biosensor model. The biological model is intentionally simplified: HER2 is represented as a diluted species transported through a lymph-node-inspired domain and captured by immobilized antibodies on a sensing surface.

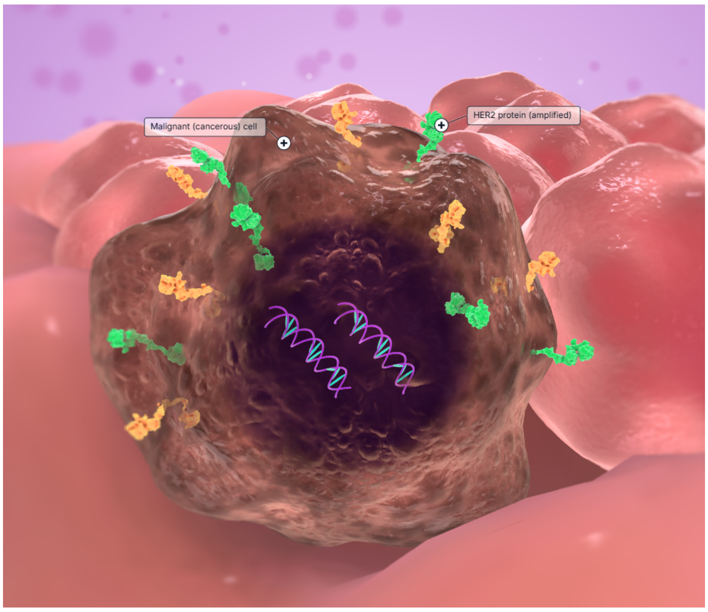

Figure 1.2: Malignant breast-cell visualization used to motivate HER2-related biomarker monitoring [5].

1.3 Conventional HER2 Testing and Limitations

Current HER2 testing methods include immunohistochemistry, fluorescence in situ hybridization, enzyme-linked immunosorbent assay, and related laboratory methods. These techniques are clinically meaningful, but they require sample handling, reagents, external instruments, and interpretation by trained personnel [2, 6, 7]. They are not designed for continuous in-body monitoring. This motivates interest in label-free electronic sensors, but such sensors face separate physical limitations, especially molecular transport, surface chemistry, electrostatic screening, and electronics noise.

1.4 Why GFET Biosensing?

Field-effect biosensors detect local electrostatic perturbations near a transistor channel. The ion-sensitive field-effect transistor established the principle that an electronic channel can respond to chemical conditions near its surface [8, 9]. Graphene is attractive for biosensing because its channel is atomically thin and surface-exposed, making it sensitive to local charge changes [10–12]. In this project, a GFET is modeled as the electrical transducer: HER2 binding changes the effective local charge environment, which is represented as a drain-source current shift ΔI_{DS}.

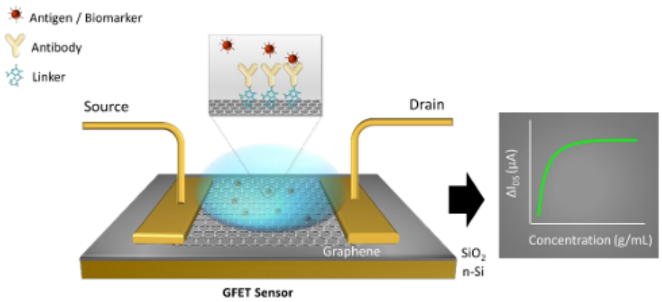

Figure 1.3: Schematic representation of GFET biosensing, where surface-bound biomolecules modulate the graphene channel current [13].

1.5 Ethical and Translational Considerations

This project is simulation-only. It does not use human data, patient samples, animal experiments, clinical testing, wet-lab validation, or fabricated devices. If a future sensor were developed for in-body use, it would require biocompatibility evaluation, safety testing, risk management, and regulatory review. ISO 10993 provides one relevant medical-device biological evaluation framework [14]. A second ethical requirement is technical honesty: simulation results should not be presented as clinical proof. For this reason, this report consistently describes the work as a simulation-based feasibility study with documented limitations.

Chapter 2

Project Definition

2.1 Problem Definition

The project studies whether HER2 can be detected electronically after accounting for the complete physical chain from antigen transport to electrical readout. The key problem is that improved molecular exposure does not automatically imply electrical detectability. The report therefore evaluates:

$$\text{transport} \rightarrow \text{binding} \rightarrow \text{transduction} \rightarrow \text{noise/screening} \rightarrow \text{detectability}.$$

The final technical question is: under what transport, binding, coupling, noise, and screening assumptions does the GFET current response exceed the detection threshold?

2.2 Solution Definition

The proposed solution is a staged modeling workflow. HER2 is introduced as a diluted species in a lymph-node-inspired transport domain. The sensor is represented by an antibody-functionalized graphene surface. Local HER2 concentration is converted into surface occupancy and bound molecule count using Langmuir binding. The bound molecule count is then converted into ΔI_{DS} using an analytical GFET current-response estimate. Debye attenuation is applied to test whether receptor-channel distance and ionic strength suppress the electrical signal below threshold.

2.3 Scope of Final Work

The final repository contains five main model stages. M1 evaluates 0D HER2-antibody binding. M2 is a diffusion-only COMSOL transport baseline. M2B is a partially anatomical prescribed-velocity convection-diffusion extension. M3 converts local sensor concentration into surface occupancy and bound molecule count. M4 maps bound molecule count to GFET current response and LOD. Experiments A to E then compare flow and diffusion, evaluate sensor placement, map detectability, evaluate Debye screening, and derive design requirements.

Compared with the midterm stage, the final work adds the M2B prescribed-velocity flow extension, M3/M4 coupling from local sensor concentration to bound molecule count and GFET current, the Experiment C detectability envelope, the Experiment D Debye-screening feasibility test, and the Experiment E design-requirement maps. These additions convert the original diffusion-and-binding concept into a full simulation-based feasibility chain while keeping the claims limited to modeled behavior.

2.4 Success Criteria and Final Interpretation

The original target was to relate HER2 concentration to ΔI_{DS} and estimate whether sub-pM detection could be predicted under selected assumptions. The final interpretation is more specific. The model predicts low-pM or sub-pM detectability only in favorable coupling and low-noise cases. It does not predict universal sub-pM detection. A result is treated as successful only if it is traceable to documented parameters, CSV outputs, figures, and verification notes. The final verification verdict is accepted with documented limitations.

Chapter 3

Related Work

3.1 ISFET and FET Biosensor Foundations

Bergveld's ion-sensitive field-effect transistor work established that an electronic channel can respond to chemical conditions near its surface [8, 9]. Later FET biosensor research extended this principle to biomolecular sensing, where binding events are converted into channel modulation [15]. This project uses the same conceptual basis but applies it to a simplified HER2-GFET feasibility model.

3.2 FET-Based Immunosensors

Immunosensors use antigen-antibody specificity to detect target molecules. A field-effect immunosensor combines selective binding with electrical transduction. The main difficulty is that the target biomolecule must reach the surface and that the binding-induced electrostatic perturbation must survive the electrolyte environment. Surface-based biosensors are also affected by reaction, diffusion, and convection limits [16].

3.3 Graphene and GFET Biosensors

Graphene-based biosensors are attractive because of high surface sensitivity and label-free operation [10–12]. However, graphene's sensitivity alone is not enough for an in-body biomarker concept. The sensing chemistry, receptor distance, ionic strength, device noise, and transport environment all affect detectability. This report uses an analytical GFET response estimate rather than a full graphene semiconductor model.

3.4 HER2 and Cancer Biomarker Detection

HER2 is clinically important for breast cancer prognosis and therapy selection [1, 2]. Conventional HER2 testing is not replaced by this project. Instead, HER2 provides a clinically meaningful target for studying whether a GFET-style transduction chain could be physically plausible under idealized and then screened assumptions.

3.5 Position of This Work

This work is positioned between biological motivation and electronic sensor design. It does not introduce a new clinical assay. Its contribution is a reproducible simulation workflow that connects lymph-node-inspired transport, antibody binding, analytical GFET current response, Debye screening, and design requirements. The central scientific thesis is that directional lymph-node-inspired flow and sensor placement improve HER2 exposure at the sensing region, but practical GFET detectability is mainly limited by electrostatic screening, receptor-channel distance, transduction coupling, binding affinity, and electronics noise.

Chapter 4

Methodology

4.1 Overall Modeling Pipeline

The model pipeline is divided into staged modules so that each physical assumption can be checked independently. The pipeline is:

$$C_{HER2} \to c_{sensor}(t) \to \theta(t) \to N_{bound}(t) \to \Delta I_{DS}(t) \to \text{detectability}.$$

This structure follows the midterm report's binding and diffusion rationale while adding final flow, placement, screening, and requirement-map analysis.

HER2 transport → sensor concentration
→ occupancy → bound molecule count → GFET current
→ noise/screening → design requirements

Figure 4.1: Final methodology pipeline used to connect transport results to electrical detectability and design requirements.

4.2 Stage M1: HER2-Antibody Binding Kinetics

HER2-antibody binding is represented by the reversible reaction

$$Ag + Ab \rightleftharpoons AgAb, \tag{4.1}$$

with reaction rate

$$r = k_f[Ag][Ab] - k_r[AgAb]. \tag{4.2}$$

The dissociation constant is

$$K_d = \frac{k_r}{k_f}. \tag{4.3}$$

For the surface occupancy calculation, the equilibrium Langmuir relation is

$$\theta = \frac{C}{K_d + C}. \tag{4.4}$$

The baseline values are $K_d = 10$ pM, $k_f = 10^7\ \mathrm{M^{-1}s^{-1}}$, and $k_r = 10^{-4}\ \mathrm{s^{-1}}$.

4.3 Stage M2: Diffusion-Only COMSOL Transport Baseline

The diffusion-only transport baseline uses COMSOL Transport of Diluted Species in a simplified cortex/medulla-inspired domain. The governing equation is

$$\frac{\partial c}{\partial t} = \nabla \cdot (D \nabla c), \tag{4.5}$$

with a heterogeneous diffusion coefficient. The medulla coefficient is parameterized by

$$D_{medulla} = r D_{cortex}, \tag{4.6}$$

where $r \in \{0.1, 0.25, 0.5, 0.75, 1.0\}$. This stage preserves the midterm diffusion-only baseline and uses it as the control case for later flow comparisons.

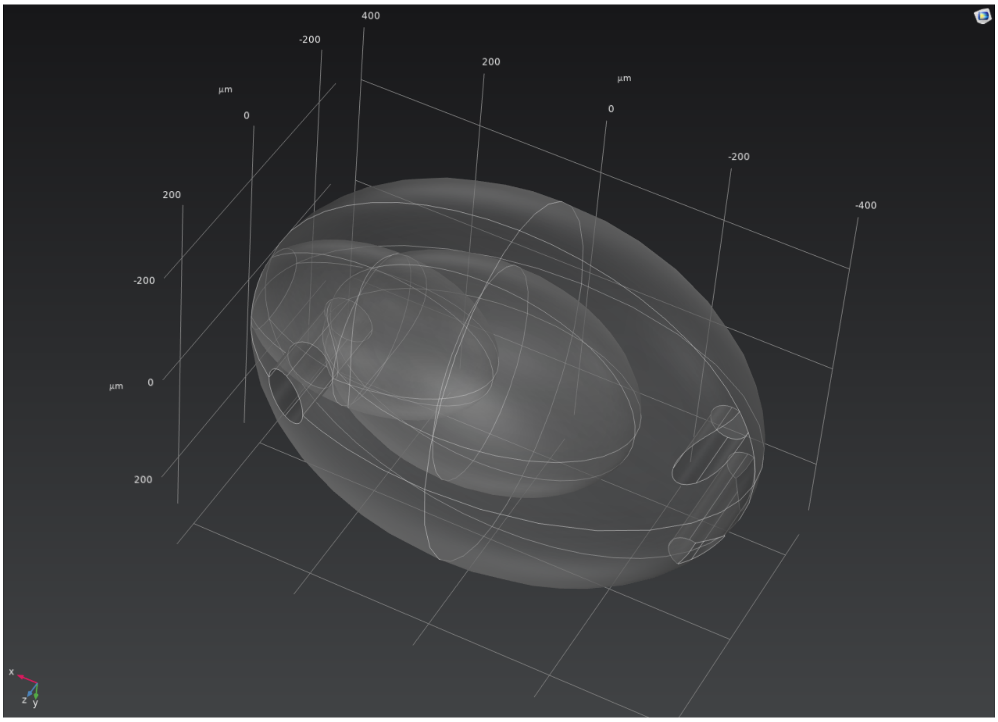

Figure 4.2: Lymph-node-inspired 3D COMSOL geometry used for the diffusion-only transport model. The outer domain represents the cortex/capsule region and the inner domain represents the medulla.

4.4 Stage M2B: Partially Anatomical Prescribed-Flow Extension

M2B is a partially anatomical prescribed-velocity convection-diffusion extension, not a full anatomical lymph-node model and not a validated Darcy-flow pressure solution. It defines four afferent inlet regions, one hilum-side efferent outlet, a capsule/no-flow marker, and local/full sensor regions. HER2 concentration is applied only to the afferent inlet regions. The efferent outlet is distinct from the afferent inlets.

The transport equation becomes

$$\frac{\partial c}{\partial t} + \mathbf{u} \cdot \nabla c = \nabla \cdot (D \nabla c), \tag{4.7}$$

where the prescribed velocity components are coupled to Transport of Diluted Species as

$$u = u_{flow}, \qquad v = v_{flow}, \qquad w = w_{flow}. \tag{4.8}$$

The velocity sweep is $v_{in} = 0$, 10^{-7}, 5×10^{-7}, and 10^{-6} m/s. The $v_{in} = 0$ case is used as a diffusion-like verification case.

4.5 Stage M3: Surface Binding and Bound Molecule Count

The local sensor concentration is converted into surface occupancy using Eq. 4.4. The bound surface density is

$$\Gamma = \Gamma_{max}\theta, \tag{4.9}$$

and the surface-bound molecule count is

$$N_{bound} = \Gamma A_{sensor} N_A, \tag{4.10}$$

where A_{sensor} is the active sensing area and N_A is Avogadro's number.

4.6 Stage M4: GFET Current Response and LOD

The analytical GFET transduction estimate is

$$\Delta I_{DS} = \frac{W}{L} e\mu V_{DS} \frac{\alpha N}{A_{eff}}, \tag{4.11}$$

where W/L is the aspect ratio, e is elementary charge, μ is carrier mobility, V_{DS} is drain-source voltage, α is effective coupling efficiency, N is bound molecule count, and A_{eff} is the effective sensing area.

The LOD is estimated as

$$LOD = \frac{3\sigma}{S}, \tag{4.12}$$

where σ is the assumed current noise floor and S is the low-concentration sensitivity [17]. In the final LOD summaries, S is fit over the low-concentration range $C = 0.5$, 1, and 10 pM. The detection criterion is

$$\Delta I_{DS} \geq 3\sigma. \tag{4.13}$$

4.7 Debye Screening Model

Electrolyte screening is modeled using the attenuation factor

$$\alpha_{eff} = \alpha_0 \exp\left(-\frac{d}{\lambda_D}\right), \tag{4.14}$$

where d is the effective receptor-channel charge distance and λ_D is the Debye length [18]. This is a Debye-screened feasibility model rather than a full electrostatic COMSOL solution.

4.8 Experiment E: Feasibility Requirement Map

Experiment E converts the Debye-screening limitation into engineering requirements. Combining Eqs. 4.11, 4.12, and 4.14 with the detection threshold gives the required coupling condition

$$\alpha_{0,req} = \frac{3\sigma A_{eff}}{(W/L)e\mu V_{DS} N_{bound}} \exp\left(\frac{d}{\lambda_D}\right). \tag{4.15}$$

The same sweep also computes maximum allowed receptor-channel distance and maximum tolerable noise floor. These values indicate which physical design changes would be required for detectability.

4.9 Verification and Reproducibility

The verification workflow checks geometry definitions, parameter consistency, nonnegative concentrations, bounded occupancy, monotonic trends, M2B convection coupling, velocity-sweep behavior, mesh sensitivity, and output-file presence. The final verdict in the verification report is accepted with documented limitations. The main reproducibility artifacts are the COMSOL Java files, solver logs, processed CSV files, plotting scripts, report figures, and verification documents.

Chapter 5

Implementation

5.1 Repository Structure

The repository is organized around the staged model. COMSOL Java files are stored under `comsol/models/`. Processed data are stored under `results/processed_csv/`. Report-ready figures are stored under `results/figures_for_report/`. Plotting and analysis scripts are stored under `scripts/`. Documentation files describe assumptions, parameters, workflow, simulation logs, verification, and limitations.

5.2 COMSOL Model Construction

The M1, M2, M2B, M3, and M4 model files are represented by Java sources and saved model files. M2 uses a meshed COMSOL Transport of Diluted Species solve for the diffusion-only baseline. M2B extends the transport model with a partially anatomical prescribed-velocity field. The M2B geometry includes afferent inlet markers, a hilum-side efferent outlet marker, and local/full sensor markers. The flow field is prescribed and coupled into the transport equation; it is not solved as a pressure-driven Darcy or Brinkman model.

5.3 Post-Processing Scripts

The processed CSV files and plots are produced from documented model parameters and sweep definitions. M3 surface binding is calculated from sensor concentration. M4 GFET response is calculated analytically from bound molecule count. Experiment E is generated from the same response model with Debye attenuation and a 3σ detection threshold.

5.4 Parameter Sweeps

The final parameter sweeps cover HER2 concentration, diffusivity ratio, prescribed inlet velocity, sensor placement, coupling efficiency, current noise floor, binding affinity, Debye length, and receptor-channel distance. Table 5.1 summarizes the main values used in the final report.

Table 5.1: Main parameters used in the final simulation campaign.

Category	**Symbol**	**Value**	**Unit**	**Use**
HER2 concentration sweep	C	0.5, 1, 10, 100, 1000	pM	Binding, transport, detectability sweeps
Baseline affinity	K_d	10	pM	M1/M3 baseline Langmuir binding
Affinity sweep	K_d	1, 10, 100	pM	Detectability and requirement maps
Forward rate constant	k_f	10^7	$M^{-1}s^{-1}$	Reversible binding assumption
Reverse rate constant	k_r	10^{-4}	s^{-1}	$k_r/k_f = 10$ pM
Cortex diffusivity	D_{cortex}	8×10^{-11}	m^2/s	Protein-scale transport baseline
Diffusivity ratio	r	0.1, 0.25, 0.5, 0.75, 1.0	none	Medulla uptake-delay sweep
M2B inlet velocity	v_{in}	0, 10^{-7}, 5×10^{-7}, 10^{-6}	m/s	Prescribed-flow sweep
Maximum binding site density	Γ_{max}	8.30×10^{-12}	mol/m^2	Effective antibody site density for M3 binding
Active sensor area	A_{sensor}	4×10^{-11}	m^2	Local sensor area for bound molecule count
Channel width	W	20	μm	GFET geometry assumption
Channel length	L	2	μm	GFET geometry assumption
Mobility	μ	0.1	$m^2/(V\ s)$	Analytical GFET response
Drain-source bias	V_{DS}	0.05	V	Analytical GFET response

Category	Symbol	Value	Unit	Use
Effective sensing area	A_{eff}	4×10^{-11}	m^2	Local GFET active area
Coupling efficiency sweep	α_0	0.01, 0.03, 0.1	none	Electrical detectability and screening
Noise floor sweep	σ	1, 5, 10, 50, 100	pA	Detection-threshold analysis
LOD fit range	C_{fit}	0.5, 1, 10	pM	Low-concentration sensitivity fit
Debye lengths	λ_D	9.6, 3.0, 0.8	nm	Low, intermediate, and PBS-like screening cases
Effective charge distance	d	0.5, 1, 2, 5, 10	nm	Debye-screened requirement map

5.5 Verification Workflow

Verification is treated as part of implementation rather than an afterthought. The workflow checks that the $v_{in} = 0$ M2B case approximates diffusion-only behavior, increasing v_{in} changes sensor exposure, directional flow creates spatial asymmetry, HER2 concentration remains nonnegative, and PBS-like screening reduces detectability. Direct COMSOL field-table export is not available for all cases, so the report identifies which quantitative tables are COMSOL-stage post-processed outputs.

Chapter 6

Results and Discussion

6.1 Binding Kinetics Result

The binding block behaves as expected. Occupancy increases monotonically with concentration and remains bounded between 0 and 1. This is an equation-level verification result rather than a biological calibration result.

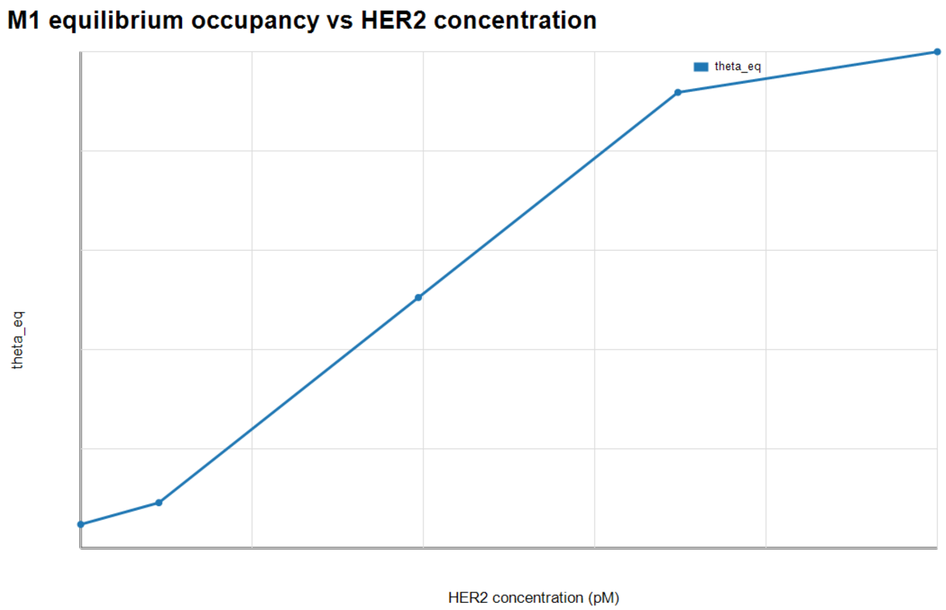

Figure 6.1: M1 equilibrium occupancy versus HER2 concentration.

6.2 Diffusion-Only Transport and Cortex-Medulla Delay

The M2 diffusion-only baseline preserves the midterm transport concept and adds a meshed COMSOL-stage implementation. The cortex/medulla-inspired geometry shows delayed medulla uptake when the medulla diffusivity is lower than the cortex diffusivity. The diffusivity-ratio sweep shows that uptake delay decreases as r approaches 1.

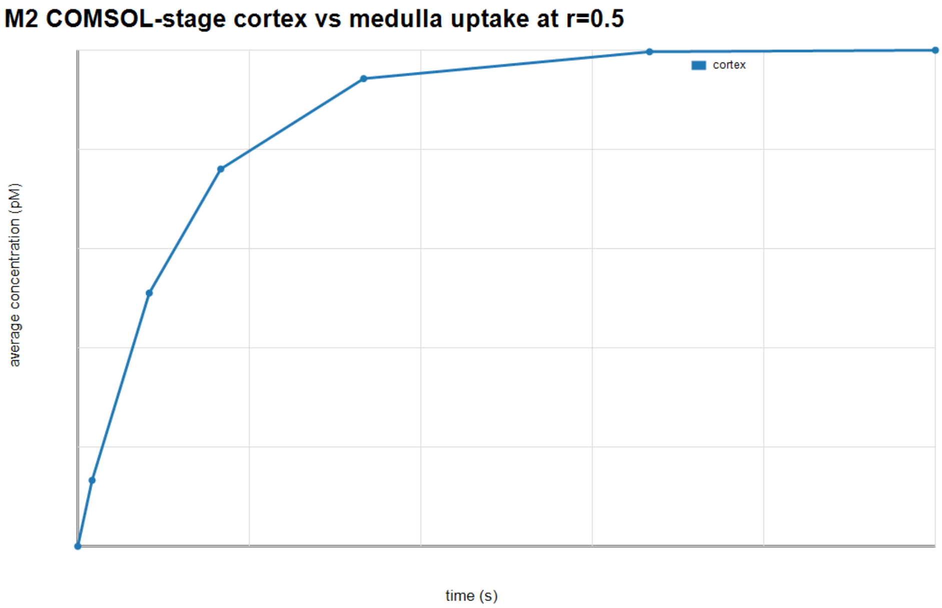

Figure 6.2: M2 cortex, medulla, and sensor concentration trends for the reference diffusion-only case.

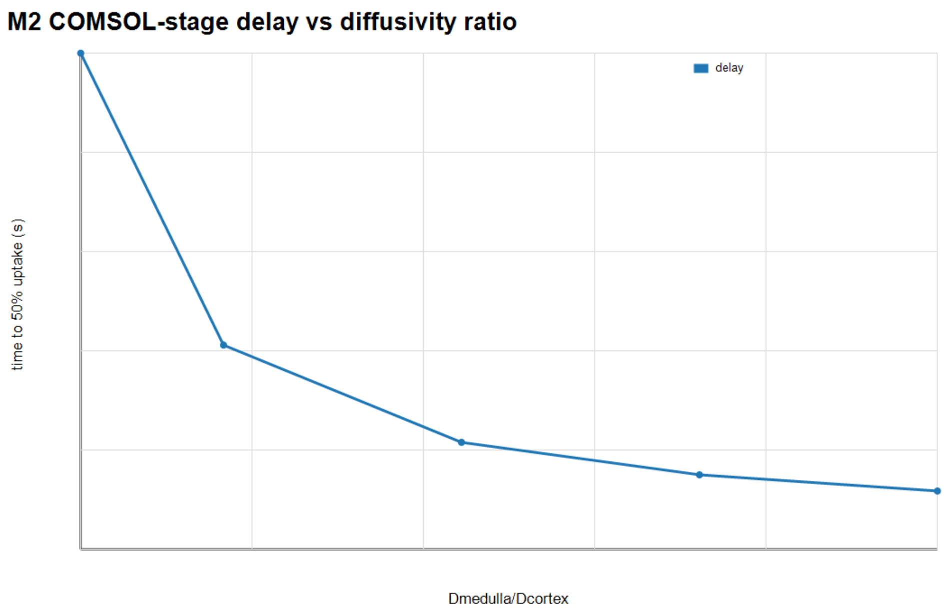

Figure 6.3: M2 medulla uptake delay versus diffusivity ratio.

6.3 Experiment A: Flow Versus Diffusion

Experiment A compares the M2 diffusion-only baseline with M2B directional-flow cases at $C = 10$ pM. This is a modeled comparison between two staged transport representations, not a perfectly matched controlled physics experiment. The cases are aligned by inlet concentration, reporting times, diffusivity ratio, and sensor-exposure metric, but M2B adds a prescribed velocity field and a partially anatomical scaffold. The comparison should therefore be read as directional sensitivity evidence rather than as an exact isolation of one physical variable.

The reference M2B flow case, $v_{in} = 5 \times 10^{-7}$ m/s, gives approximately 8.73 pM sensor concentration at 6000 s. The M2 diffusion-only baseline gives approximately 4.98 pM at the same time. This is approximately a 1.75x increase in late-time sensor exposure. The time-to-50% exposure decreases from about 6038 s to about 1839 s. The result supports the conclusion that prescribed lymph-node-inspired flow can improve sensor exposure under the modeled assumptions.

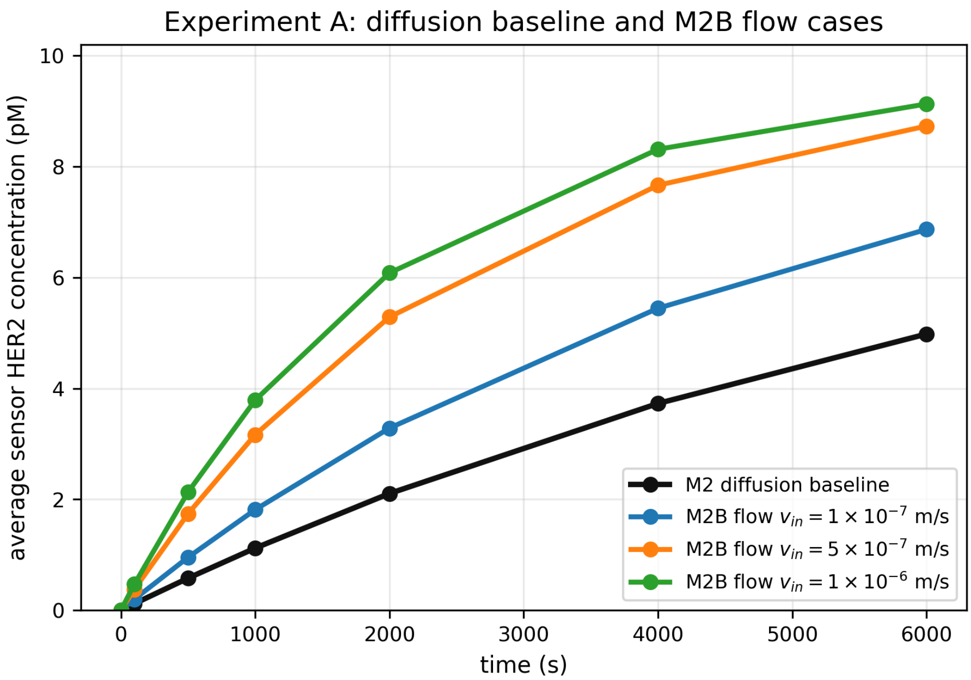

Figure 6.4: Experiment A sensor exposure for the M2 diffusion baseline and M2B prescribed-flow cases at $v_{in} = 1 \times 10^{-7}$, 5×10^{-7}, and 1×10^{-6} m/s.

6.4 Experiment B: Sensor Placement Sensitivity

Experiment B evaluates three placements: subcapsular/cortical, cortex-medulla transition, and medulla/hilum-side. Sensor placement affects timing and current response. In the modeled sweep, the subcapsular/cortical placement is fastest and strongest under the reference flow case. Under directional flow, late-time exposure becomes high across all tested placements, while placement still changes arrival time and pathway exposure. The placement sweep is post-processed over the transport model, not a full remeshed COMSOL relocation study.

The current-response range across placements is approximately 559.9 to 600.5 pA under flow and 348.2 to 533.3 pA under diffusion-only transport. Therefore, the late-time current spread is smaller under flow than under diffusion-only transport, even though placement still affects response timing.

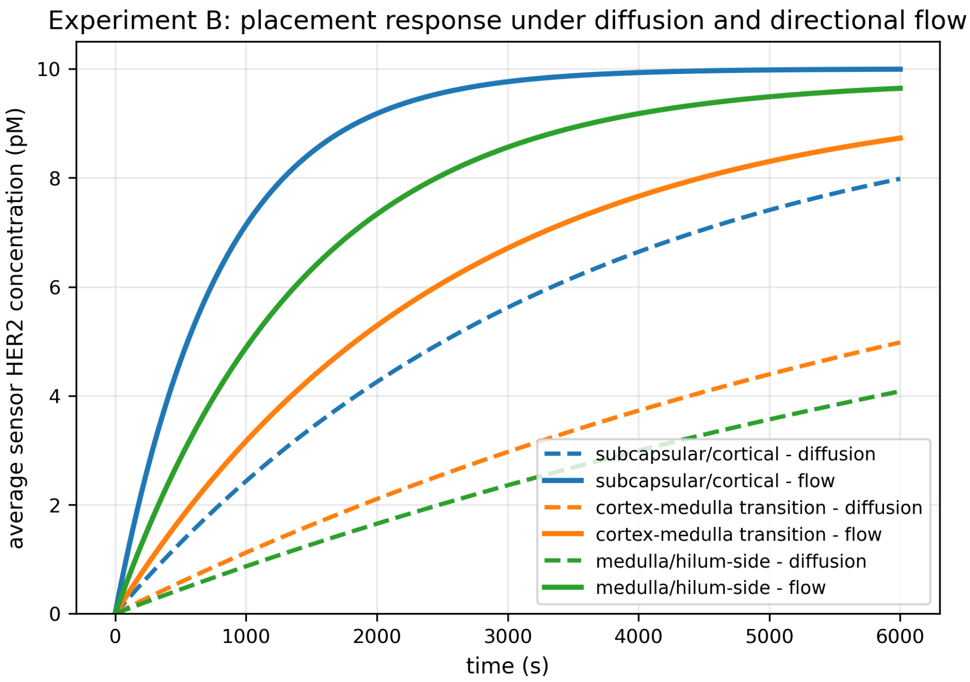

Figure 6.5: Experiment B concentration exposure for subcapsular/cortical, cortex-medulla transition, and medulla/hilum-side placements under diffusion-only and directional-flow transport.

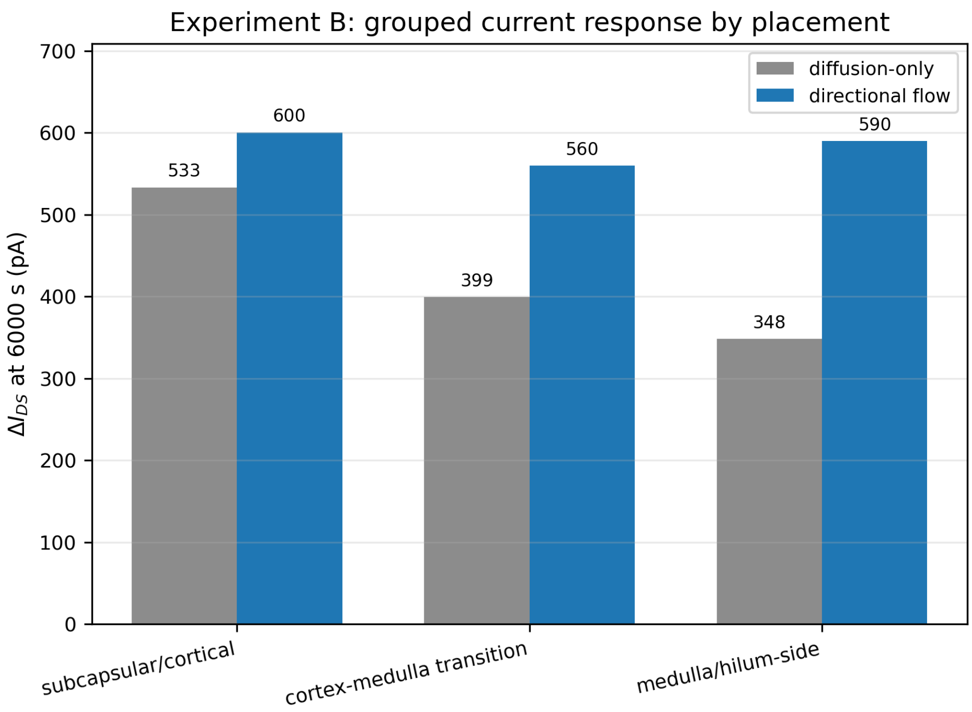

Figure 6.6: Experiment B grouped GFET current response at 6000 s comparing diffusion-only and directional-flow transport for the three sensor placements.

6.5 Surface Binding and GFET Current Response

Surface binding converts sensor concentration into bound molecule count using the documented Γ_{max} and local sensor-area assumptions. The local GFET-scale sensing area produces the count used for current response. The M4 current response increases with HER2 concentration and coupling efficiency. Under the current assumptions, the $\alpha = 0.03$, 10 pA case reaches a simulated LOD of approximately 0.78 pM using the 0.5 to 10 pM fit range. This value is conditional and should not be generalized to all noise, coupling, or screening cases.

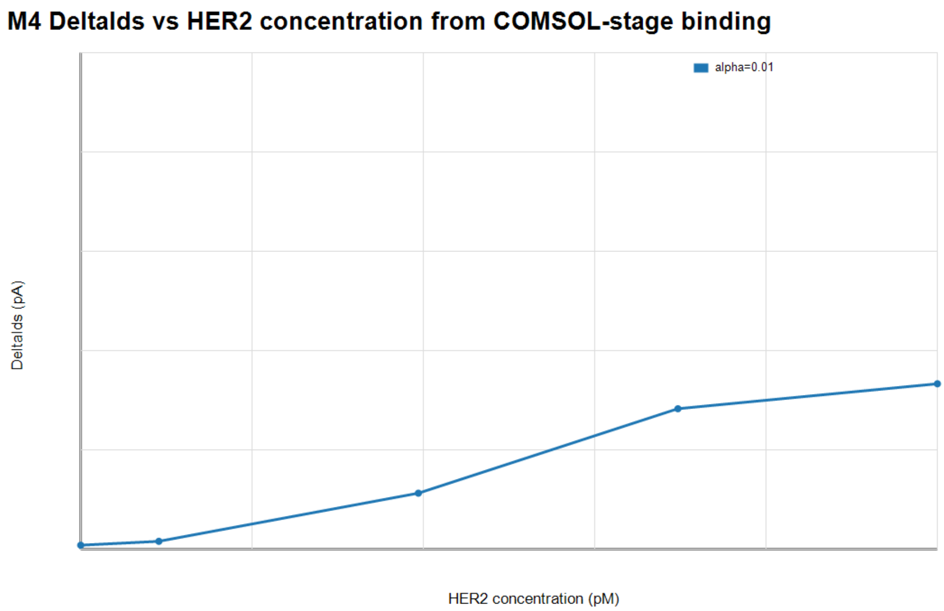

Figure 6.7: M4 analytical GFET current response versus HER2 concentration.

6.6 Experiment C: Detectability Envelope

Experiment C sweeps HER2 concentration, coupling efficiency, current noise floor, and binding affinity. Detectability is mixed: 99 parameter rows are detectable and 36 rows fail the 3σ threshold. Detection is favored by stronger coupling, lower noise, tighter binding, and higher concentration. Low concentration, weak coupling, high noise, or poor affinity can cause failure. The correct conclusion is conditional detectability, not universal sub-pM detection.

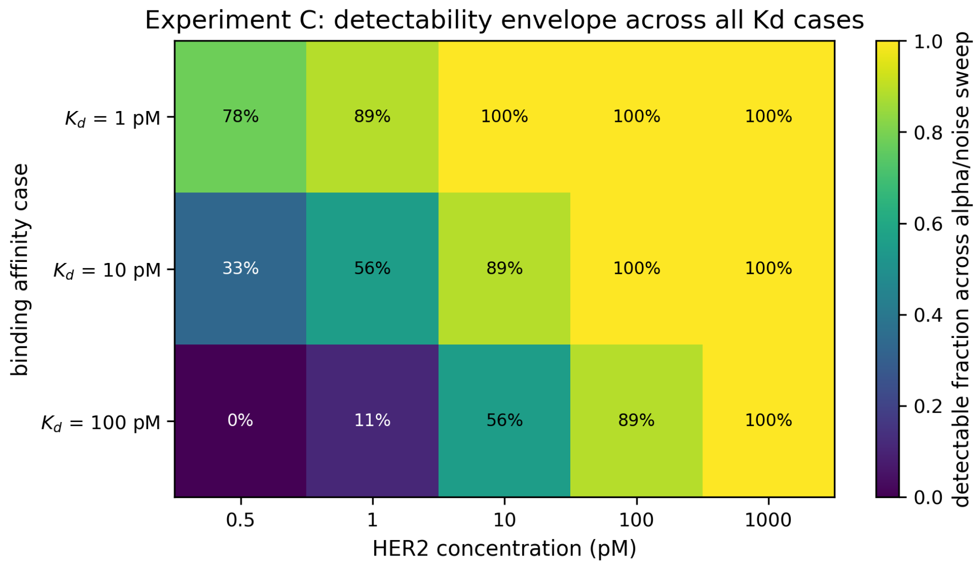

Figure 6.8: Experiment C detectability envelope across all modeled HER2 concentration and K_d cases, with each cell showing the detectable fraction over the coupling/noise sweep.

6.7 Experiment D: Debye Screening Feasibility

Experiment D applies the Debye-screened coupling model in Eq. 4.14. PBS-like physiological screening strongly suppresses the GFET signal. For PBS-like $\lambda_D = 0.8$ nm and $d \geq 5$ nm, 0 of 60 screened cases remain detectable. This is the main negative feasibility result. Transport improvement alone cannot overcome screening when the effective charge is too far from the graphene channel.

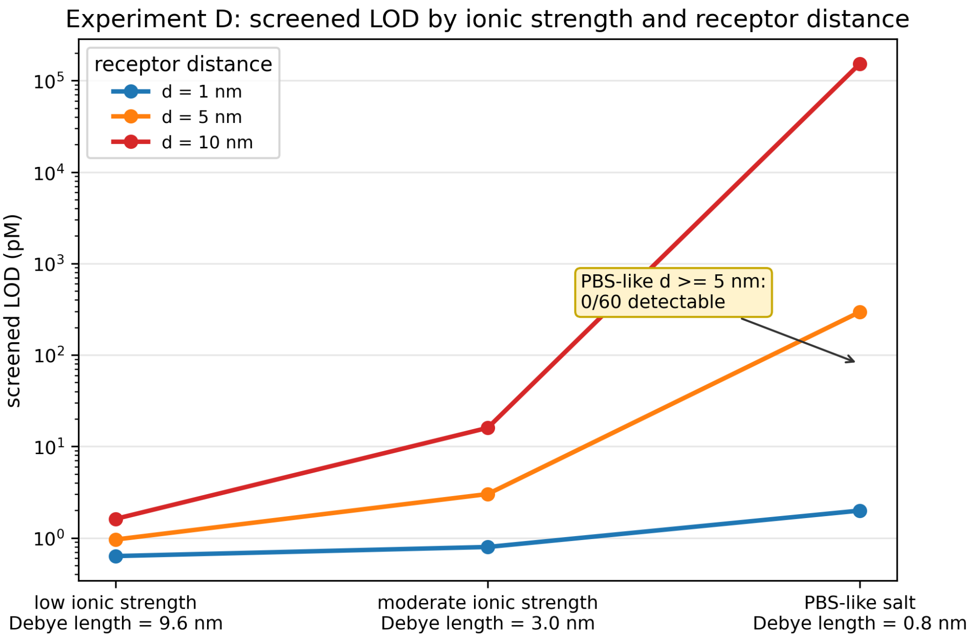

Figure 6.9: Experiment D screened LOD versus ionic-strength case and effective receptor-channel distance for the reference $\alpha_0 = 0.03$, 10 pA, $C = 10$ pM case.

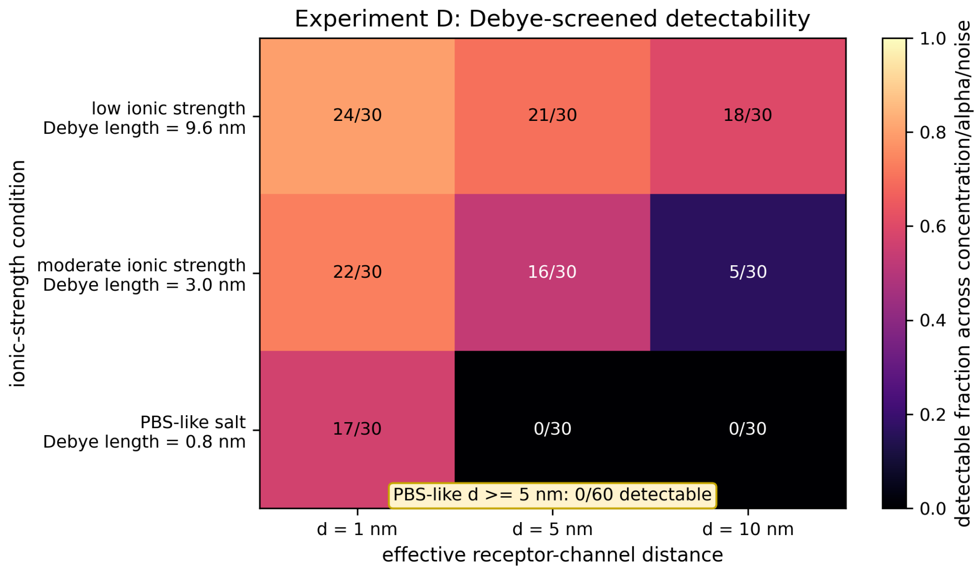

Figure 6.10: Experiment D screened detectability map under Debye attenuation assumptions, including the PBS-like $d \geq 5$ nm condition where 0 of 60 cases remain detectable.

6.8 Experiment E: Design Requirement Map

Experiment E converts the Debye-screening result into explicit engineering constraints. Under PBS-like screening, detection is not robust beyond approximately 1 nm effective receptor-channel distance for the baseline case. For $K_d = 10$ pM, 10 pA noise, and PBS-like screening at $d = 1$ nm, the required α_0 is approximately 0.0326 at 1 pM and 0.00561 at 10 pM. At $d = 2$ nm, the required α_0 increases to approximately 0.114 at 1 pM and 0.0196 at 10 pM.

The design implication is direct: feasibility requires short receptor distance, strong transduction, low noise, favorable binding affinity, and control of screening. The model does not prove that an in-body GFET-HER2 detector is clinically feasible. It identifies the physical and electronic conditions that would have to be met.

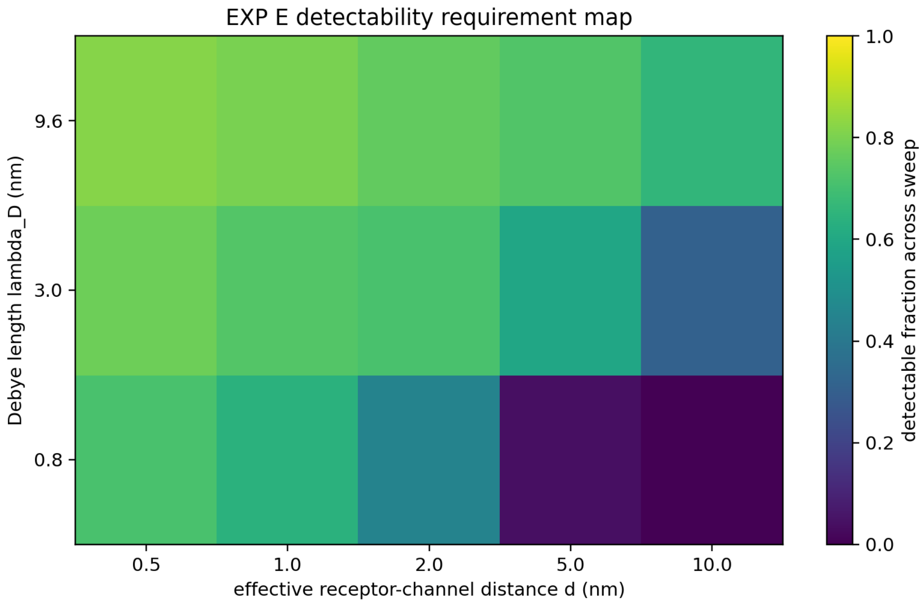

Figure 6.11: Experiment E detectability requirement map showing detectable sweep fraction versus Debye length and effective receptor-channel distance.

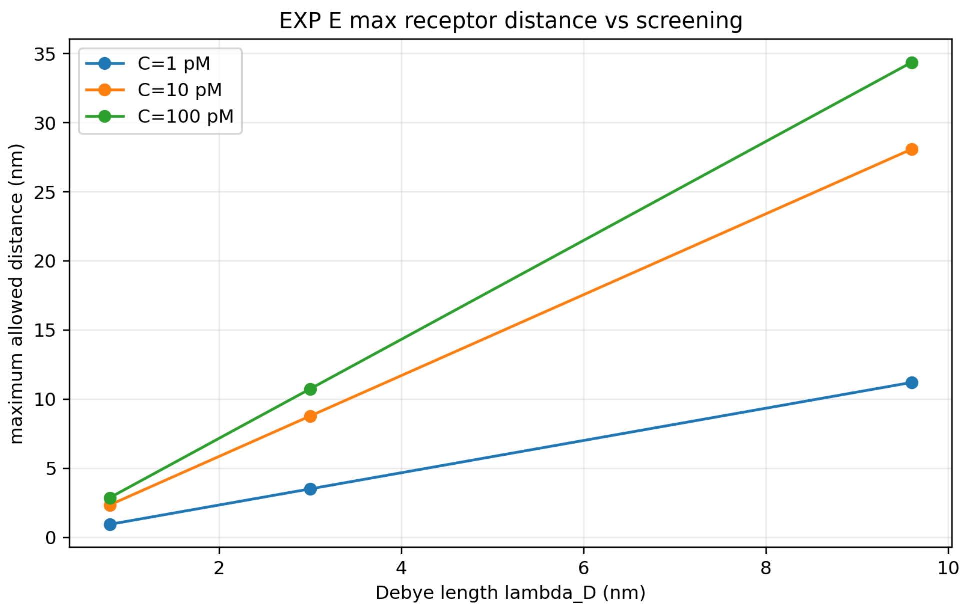

Figure 6.12: Experiment E maximum allowed receptor-channel distance versus Debye length for selected concentration cases.

Table 6.1: Representative Experiment E design requirements for $K_d = 10$ pM, $C = 10$ pM, $\alpha_0 = 0.03$ where applicable, and a 10 pA reference noise floor.

Ionic condition	**Distance**	**Required α_0**	**Max. allowed noise**	**Conclusion**
Low ionic strength, $\lambda_D = 9.6$ nm	0.5 nm	0.00169	177.15 pA	Detectable under favorable screening
Intermediate screening, $\lambda_D = 3.0$ nm	2 nm	0.00313	95.81 pA	Detectable with moderate margin
PBS-like screening, $\lambda_D = 0.8$ nm	1 nm	0.00561	53.47 pA	Detectable at 10 pM, but distance-sensitive
PBS-like screening, $\lambda_D = 0.8$ nm	2 nm	0.0196	15.32 pA	Detectable only with stronger coupling or low noise
PBS-like screening, $\lambda_D = 0.8$ nm	5 nm	0.833	0.360 pA	Not robust for the baseline design

6.9 Final Scientific Claims

The positive findings are that directional flow improves sensor exposure, sensor placement changes response timing/current, and favorable coupling/noise/affinity combinations can make the analytical current response detectable. The negative findings are that detection is not universal, PBS-like screening can eliminate detectability for longer receptor distances, and transport improvement alone does not guarantee electrical detectability. The main interpretation is that transport can be improved, but in-body GFET-HER2 detection is mainly constrained by electrostatic screening, receptor distance, transduction coupling, and electronics noise.

Table 6.2: Final scientific claims supported by the simulation campaign.

Claim	**Supporting result**	**Fig.**	**Assumption**	**Limitation**	**Conf.**
HER2-antibody binding follows bounded saturation behavior across the modeled concentration range.	M1 occupancy increases monotonically with HER2 concentration and remains within [0, 1].	Fig. 6.1	Reversible Langmuir-type binding with baseline $K_d = 10$ pM.	No wet-lab binding calibration was performed.	High
Directional lymph-node-inspired flow changes sensor exposure compared with diffusion-only transport.	At $C = 10$ pM, $v_{in} = 5 \times 10^{-7}$ m/s gives 8.73 pM at 6000 s versus 4.98 pM for diffusion-only; time-to-50% exposure decreases from 6038 s to 1839 s.	Fig. 6.4	Prescribed velocity is coupled into TDS convection and compared by common reported metrics.	M2 versus M2B is a modeled comparison, not a perfectly matched controlled physics experiment.	Medium
Sensor placement affects response timing and current response.	Subcapsular/cortical placement is fastest and strongest under reference flow; late-time current range is 559.9 to 600.5 pA under flow and 348.2 to 533.3 pA under diffusion-only transport.	Figs. 6.5 and 6.6	Three placement regions are post-processed over the same transport model.	Placement sweep is not a full remeshed COMSOL sensor relocation study.	Medium
Sub-pM or low-pM detectability is predicted only under favorable coupling, low-noise, and binding assumptions.	Experiment C gives 99 detectable rows and 36 failing rows across concentration, α_0, noise, and K_d sweeps.	Fig. 6.8	Analytical GFET response with $\Delta I_{DS} \geq 3\sigma$.	Graphene electronics and surface chemistry are simplified.	Medium
PBS-like Debye screening strongly reduces effective GFET signal for long receptor distances.	For PBS-like $\lambda_D = 0.8$ nm and $d \geq 5$ nm, 0 of 60 screened cases remain detectable.	Figs. 6.9 and 6.10	Screened coupling model.	Debye lengths and distances are approximate feasibility assumptions.	Medium
Direct in-body HER2 detection likely requires short receptor distance, low-noise electronics, stronger transduction, or local screening mitigation.	Transport and placement improve antigen exposure, but detection fails under weak coupling, high noise, poor affinity, or PBS-like screening at longer distances.	Table 6.1	Synthesis of M2B exposure, M3 binding, M4 response, and Debye attenuation.	Simulation-based design conclusion, not clinical validation.	High

Claim	Supporting result	Fig.	Assumption	Limitation	Conf.
Feasible GFET-HER2 detection requires explicit design conditions on receptor distance, coupling, screening, and electronics noise.	Experiment E maps required α_0, maximum allowed receptor-channel distance, and maximum allowed noise floor. For $K_d = 10$ pM, 10 pA noise, and PBS-like screening at 1 nm, required α_0 is 0.0326 at 1 pM and 0.00561 at 10 pM.	Fig. 6.11	M3/M4 response with screened coupling.	Simulation-based requirement analysis, not fabricated-device performance.	Medium

6.10 Compact Final Summary

Table 6.3: Compact final summary of the simulation results and interpretation.

Question	Result	Interpretation
What was calculated?	A staged chain from HER2 transport to sensor concentration, occupancy, bound molecule count, GFET current, noise/screening, and design requirements.	The project is a simulation-based feasibility study, not a fabricated or clinically validated detector.
Does flow change sensor exposure?	At 10 pM, the $v_{in} = 5 \times 10^{-7}$ m/s case gives 8.73 pM at 6000 s versus 4.98 pM for diffusion-only transport.	Directional prescribed flow improves exposure and reduces the time-to-50% exposure under the modeled assumptions.
Does sensor placement matter?	Subcapsular/cortical placement is fastest and strongest under reference flow; late-time flow exposure remains high across all three tested placements.	Placement affects response timing and pathway exposure, but flow reduces the late-time current spread compared with diffusion-only transport.
When is current detectable?	Experiment C gives 99 detectable and 36 failing rows across concentration, coupling, noise, and K_d cases.	Detectability is conditional and depends on coupling, noise, affinity, and concentration; it is not universal sub-pM detection.
What is the main negative result?	PBS-like $\lambda_D = 0.8$ nm screening with $d \geq 5$ nm gives 0 of 60 detectable screened cases.	Electrostatic screening and receptor-channel distance can dominate the feasibility limit even when transport exposure improves.
What would need to improve?	Experiment E points to short effective receptor distance, strong transduction, low noise, favorable binding, and screening mitigation.	Future work should focus on surface chemistry, electronic noise, validated flow physics, and device characterization.

Chapter 7

Limitations

7.1 Simulation-Only Scope

No wet-lab validation, fabricated GFET device, clinical validation, animal experiment, or patient-data study is included. The results are simulation-based design conclusions.

7.2 COMSOL Export and Post-Processing Limitations

Direct COMSOL field-table export is not available for all cases. Some quantitative tables are post-processed from documented model assumptions and parameter sweeps. These tables are useful for final synthesis, but they should not be described as measured field-probe exports for every case.

7.3 Prescribed-Flow Limitation of M2B

M2B is a partially anatomical prescribed-velocity convection-diffusion extension, not a full anatomical lymph-node model and not a validated Darcy-flow pressure solution. It is suitable for comparing a controlled directional-flow scenario with diffusion-only transport, but it is not a validated physiological lymph-node flow model.

7.4 Analytical GFET Response Limitation

The GFET response is analytical. It does not solve a full graphene semiconductor transport model, contact effects, hysteresis, fabrication variability, electrolyte-gate effects, or noise spectra. The current response should therefore be interpreted as a transduction estimate.

7.5 Debye Screening Approximation

Debye screening is modeled through exponential attenuation. The model does not solve full electrostatics in COMSOL. The Debye-screened analysis is useful because it shows that receptor distance and ionic strength can dominate detectability, but the exact thresholds require experimental device characterization.

7.6 Clinical and Biocompatibility Limitations

The project does not include clinical validation, implant packaging, long-term stability, immune response, fouling, toxicity, sterilization, or biocompatibility testing. Any future in-body device would require extensive safety and regulatory work.

Chapter 8

Conclusion and Future Work

8.1 Main Conclusion

The project does not prove that an in-body GFET-HER2 detector is clinically feasible. Instead, it identifies the physical and electronic conditions required for feasibility. Directional transport and placement improve antigen exposure, but the dominant limitations are Debye screening, receptor-channel distance, coupling efficiency, and electronics noise.

8.2 Engineering Requirements for Feasibility

The final design requirement map indicates that a feasible GFET-HER2 design would need a short effective receptor-channel distance, strong effective transduction coupling, low current noise floor, favorable binding affinity, and mitigation of physiological ionic screening. For the baseline PBS-like case, detection is not robust beyond approximately 1 nm effective receptor-channel distance. These values provide design requirements rather than experimental performance guarantees.

8.3 Future Work

Future work should validate the binding and GFET parameters experimentally, replace prescribed flow with a validated Darcy or Brinkman pressure-flow model, implement direct COMSOL field-table export for all cases, test receptor chemistries that reduce effective charge distance, characterize device noise, and evaluate biocompatibility and fouling. The next technical step is not to claim clinical readiness, but to test whether surface chemistry and electronics can meet the requirements identified by Experiment E.

Appendix A

Key Equations

The main equations used in the report are collected here for reproducibility:

$$\theta = \frac{C}{K_d + C}, \tag{A.1}$$

$$N_{bound} = \Gamma A_{sensor} N_A, \tag{A.2}$$

$$\Delta I_{DS} = \frac{W}{L} e\mu V_{DS} \frac{\alpha N}{A_{eff}}, \tag{A.3}$$

$$LOD = \frac{3\sigma}{S}, \tag{A.4}$$

$$\alpha_{eff} = \alpha_0 \exp\left(-\frac{d}{\lambda_D}\right), \tag{A.5}$$

$$\Delta I_{DS} \geq 3\sigma, \tag{A.6}$$

$$\alpha_{0,req} = \frac{3\sigma A_{eff}}{(W/L) e\mu V_{DS} N_{bound}} \exp\left(\frac{d}{\lambda_D}\right). \tag{A.7}$$

Appendix B

Parameter Tables

The final parameter table is provided in Table 5.1. SI units are used internally in calculations, while pM, pA, and nm are used in the discussion for readability.

Appendix C

Verification Summary

The verification report confirms bounded binding occupancy, nonnegative concentration outputs, velocity coupling in M2B, diffusion-like behavior for $v_{in} = 0$, increased sensor exposure with increasing prescribed velocity, spatial asymmetry under directional flow, mesh sensitivity checks, and export-file presence. The final verification verdict is accepted with documented limitations.

Appendix D

Repository and Reproducibility Notes

The report is supported by the following repository components:

- `docs/model_assumptions.md` and `docs/parameter_table.md` for assumptions and parameters;
- `docs/comsol_workflow.md` for workflow evidence;
- `docs/simulation_log.md` for run history;
- `docs/verification_report.md` for verification evidence;
- `results/processed_csv/` for final numerical summaries;
- `results/figures_for_report/` for report figures;
- `scripts/` for post-processing and plotting scripts;
- `report/references.bib` for IEEE-style citations.

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